

UAV Path Planning for Breaking Through Jamming Zones Based on Artificial Potential Field

YingJie Liu*
Early Warning Academy
Wuhan, Hubei, China
461686884@qq.com

Quanmin Niu
Early Warning Academy
Wuhan, Hubei, China
312666354@qq.com

Abstract

This paper presents a novel path planning approach to address the vulnerability of UAV satellite navigation systems under suppressive jamming during offensive operations. Specifically, an improved Artificial Potential Field (APF) algorithm is developed to navigate UAVs through denial-of-service environments. The study incorporates a comprehensive threat model within a constructed mission scenario. A key innovation involves setting virtual obstacle points to dynamically optimize the potential field forces, thereby enabling the fixed-wing UAV to evade jamming threats while maintaining flight constraints. Integrating the APF strategy with an Inertial Navigation System (INS), our simulations demonstrate robust penetration capabilities. Through extensive Monte Carlo analysis concerning mission duration and weight coefficients, the proposed method validates its efficacy, exhibiting superior survival and mission completion rates under varying conditions.

CCS Concepts

• **Applied computing** → Computing in other domains; Military.

Keywords

Unmanned Aerial Vehicles, Artificial potential field, Performance constraints, jamming

ACM Reference Format:

YingJie Liu and Quanmin Niu. 2026. UAV Path Planning for Breaking Through Jamming Zones Based on Artificial Potential Field. In *2026 3rd International Conference on Autonomous Driving and Intelligent Sensing Technology (ADIST 2026)*, February 06–08, 2026, Wuhan, China. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3807211.3807220>

1 Introduction

UAVs have emerged as critical assets in military operations, transportation systems, and public administration, driven by significant advances in aviation technology and weapons systems[1]. The integration of UAVs into both offensive and defensive military capabilities represents a fundamental shift in contemporary military doctrine and future force development[2-4].

In the domain of counter-UAV operations, current research predominantly centers on drone swarm warfare tactics, with an emphasis on optimizing cost-effectiveness ratios[5]. Contemporary

counter-UAV strategies encompass both kinetic (hard-kill) and non-kinetic (soft-kill) approaches, with satellite navigation interference emerging as a particularly effective soft-kill methodology[6][7].

To ensure unmanned aerial vehicles (UAVs) can effectively reach their designated targets and accomplish assigned missions, researchers have developed various obstacle avoidance path planning algorithms. These include artificial potential field methods[8-10], particle swarm optimization[11][12], and ant colony algorithms[13][14]. Zhao et al.[15] proposed a hybrid path planning algorithm combining Selfish Herd Optimizer (SHO) and Particle Swarm Optimizer (PSO) to address radar and artillery threats, demonstrating superior avoidance capabilities when threat ranges are known. Giuseppe et al.[16] developed an enhanced energy-efficient variant of the A* pathfinding algorithm, capable of identifying collision-free routes in complex environments with unknown obstacles. For three-dimensional obstacle avoidance in fixed-wing UAVs, Lin et al.[17] introduced the Fast Geometric Avoidance (FGA) algorithm, which effectively navigates multiple static and dynamic obstacles while maintaining UAV performance constraints and original trajectory recovery. Wang et al. [18] enhanced the artificial potential field algorithm by incorporating UAV physical constraints and optimizing solutions for local minima and excessive repulsive forces in trajectory planning. Sun et al.[19] investigated dynamic obstacle avoidance using a danger index-based artificial potential field approach, which considers both distance factors and relative velocity relationships between UAVs and obstacles.

In combat scenarios, obstacles typically manifest as "threat circles" rather than physical barriers. When UAVs enter enemy engagement zones, they encounter various anti-aircraft weapons and electronic warfare systems. Unlike physical obstacles, these threats do not cause direct collision but require tactical flight adjustments within the threat zone to ensure successful mission completion.

While path planning for UAVs in electronic warfare environments has gained strategic importance, relevant research remains limited. Mah[20] developed an algorithm optimizing UAV relay trajectories by calculating heading angles to maximize receiver signal-to-noise ratio. Peng [21] introduced an enhanced Q-Learning algorithm incorporating multi-parameter planning through reinforcement learning, integrating cost factors to optimize UAV mobility and communication performance. However, this approach's reliance on communication makes it vulnerable to high-power enemy jamming.

This paper presents a penetration model for fixed-wing UAVs with inertial navigation systems, utilizing an artificial potential field algorithm to navigate through electronic warfare environments. The model addresses the vulnerability of UAVs to electronic interference within penetrable threat spaces. Virtual interference zones are established based on survival probability and mission

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. ADIST 2026, Wuhan, China

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2216-5/2026/02

<https://doi.org/10.1145/3807211.3807220>

criticality, incorporating virtual obstacle points into the artificial potential field framework. The model employs attractive and repulsive potential field functions to enable dynamic flight path optimization. Through analysis of UAV motion characteristics under both ideal and constrained trajectories within interference zones, the approach facilitates adaptive penetration strategies that balance mission duration, survival probability, and mission success rate.

2 UAV Penetration Model

2.1 Drone Mission Completion Rate

The state variables of the UAV can be expressed as:

$$U(t) = [v(t), q(t), s(t)]^T \quad (1)$$

where $v(t)$, $q(t)$ and $s(t)$ represent the current velocity, position, and survival probability of the UAV respectively, the state equation of the UAV can be expressed as:

$$\begin{cases} v(t+1) = v(t) + a(t) \\ q(t+1) = q(t) + v(t+1) \\ s(t+1) = s(t) \cdot A(t) \end{cases} \quad (2)$$

where $A(t)$ represents the change in survival rate of the UAV from time t to $t+1$, which is related to the distance between the UAV and the target point. The vector $a(t)$ represents the acceleration of the UAV at time t . If the survival probability of the UAV falls below a certain threshold, the UAV is considered destroyed, resulting in mission failure. Currently, only considering the survival rate changes under distance-based situations, it can be expressed as

$$A(t) = 1 - e^{-lD/R} \quad (3)$$

where D represents the distance between the UAV and the interference source, R represents the maximum interference radius of the interference source and l is the defense coefficient, which indicates the defense effectiveness at the current target point.

To complete its mission, a UAV must not only maintain a survival rate above the survival threshold but also have sufficient attack time after reaching the attack range. Therefore, the mission completion rate of the UAV can be expressed as

$$c = \begin{cases} \frac{T-T_a}{T} \times 100\% & s(t) \geq S_T, D \leq R_s \\ 0 & s(t) < S_T, D > R_s \end{cases} \quad (4)$$

where c denotes the task completion rate; T represents the total action time allocated to the UAV from the starting point; T_a indicates the time when the mission target enters the UAV's attack range; S_T represents the UAV's survival threshold and R_s represents the attack range of the UAV.

If the UAV reaches the target range without being destroyed, it can complete the attack mission; however, if the UAV is destroyed before reaching the target range, the mission is considered a failure.

2.2 Interference Zone

When an UAV is en route to its mission target and its velocity vector points directly toward a jamming source, it must execute evasive maneuvers within the jamming zone to avoid prolonged interference that would prevent mission completion.

When the jamming zone is known, a UAV can select the optimal approach angle to evade its threat. However, in most cases, UAVs

cannot ascertain the specific range of the jamming zone when initiating an attack. Therefore, it is necessary to predict the jamming zone in advance and select an appropriate flight path to successfully complete the mission.

When an early warning radar detects a UAV, electronic jamming equipment directs electronic interference toward the UAV's specific direction. At this point, the attacking party cannot determine the exact effective range of the jamming. Assuming there exists a real jamming zone Ω with a jamming beam width of θ , a virtual jamming zone Ω' it is proposed to enable evasion, with an angle φ between the real and virtual jamming zones. Θ represents the overlapping region between the real and virtual jamming zones, as shown in Figure 1. The smaller the φ , the larger the Θ region becomes, indicating that the virtual interference zone approaches the actual interference zone more closely.

Let the axis of symmetry of Ω' coincide with the line connecting the UAV at coordinates $q(x, y)$ and the jamming source at coordinates $q_g(x_g, y_g)$. The slope angle ϑ of this axis can be expressed as

$$\vartheta = \arctan \frac{y - y_g}{x - x_g} \quad (5)$$

Consequently, the slopes k_1 and k_2 of the upper and lower boundaries of the virtual jamming zone can be expressed as

$$\begin{cases} k_1 = \tan\left(\vartheta + \frac{\theta}{2}\right) \\ k_2 = \tan\left(\vartheta - \frac{\theta}{2}\right) \end{cases} \quad (6)$$

3 Potential Field-Based Path Planning

The artificial potential field algorithm is a classic path planning algorithm. This algorithm simulates the relationship of the electric potential field between positive and negative charges, enabling the controlled object to calculate the potential energy at each position based on obstacles and targets in the scene while avoiding obstacles. It allows the drone to move toward the target along the negative gradient direction, which is the direction of decreasing potential energy.

3.1 Artificial Potential Field Algorithm

The repulsive charge will drive the UAV to move toward the target point, as shown in Figure 2. Assuming the UAV is a particle moving under the resultant force F_r , which is the vector sum of the attractive force F_{att} from the target point and the repulsive force F_{rep} from obstacles.

The traditional potential field function is described as:

$$U_{att}(q) = \frac{1}{2} \xi \sigma^2(q, q_g) \quad (7)$$

$$U_{rep}(q) = \begin{cases} \frac{1}{2} \eta \left(\frac{1}{\rho(q, q_o)} - \frac{1}{\rho_0} \right)^2 & \rho(q, q_o) \leq \rho_0 \\ 0 & \rho(q, q_o) > \rho_0 \end{cases} \quad (8)$$

In this context, $U_{att}(q)$ and $U_{rep}(q)$ represent the potential energy of the attractive field and the repulsive field, respectively; $\|\cdot\|_2$ denotes the Euclidean norm; $\sigma(q, q_g) = \|q - q_g\|_2$ represents the distance between the drone and the target point; $q_o(x_o, y_o)$ represents the coordinates of the obstacle; $\rho(q, q_o) = \|q - q_o\|_2$ represents the distance between the drone and the obstacle; ξ and η denote the attractive coefficient and the repulsive coefficient, respectively.

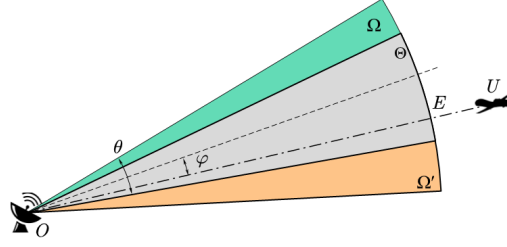


Figure 1: Schematic Diagram of Real and Virtual Interference Zones

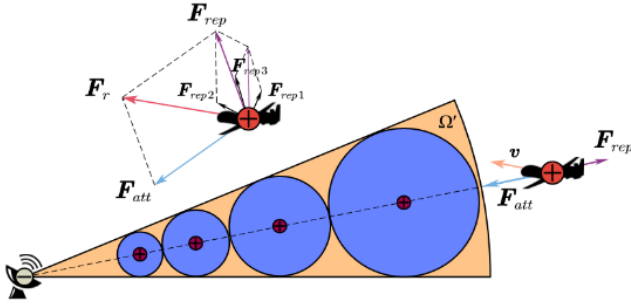


Figure 2: The force field acting on the drone

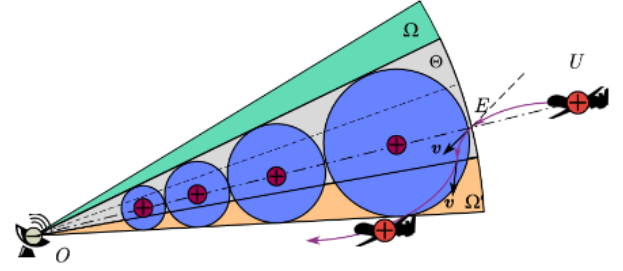


Figure 3: Flight path traversal

3.2 Improved Artificial Potential Field Algorithm

During the drone's flight toward its target, since only the target point can serve as an attractive force for the drone and it cannot obtain repulsive forces from other obstacles, additional obstacles need to be set up to control the drone's avoidance path.

When using traditional potential field functions, there is no distinction in how the UAV's current position affects its behavior when entering or leaving interference zones. After entering an interference zone, due to excessive attractive force, the UAV continues to fly toward the target point and cannot exit the interference zone. Additionally, due to the weak repulsive force, while it can generate enough repulsion to prevent direct collision with obstacles, it cannot guarantee that the UAV will truly escape the threat of the interference zone. This contradicts the current mission requirements.

To avoid the aforementioned problems while ensuring the UAV can select the optimal evasion method to exit the interference zone, when the UAV enters the actual interference zone with velocity and position as shown in Figure 3, it will exit the interference zone by breakthrough mode along the shortest distance under the influence of potential field forces. When entering with velocity and position as shown in Figure 4, the UAV will exit the interference zone in evasion mode under the influence of potential field forces, and

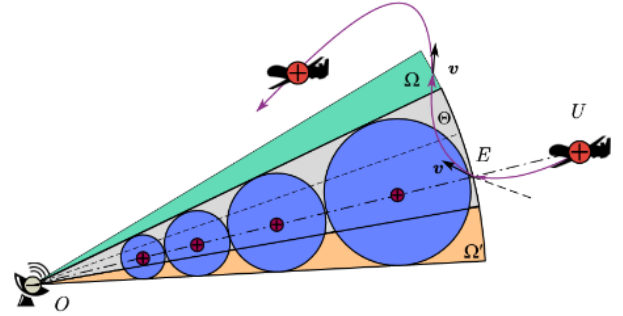


Figure 4: Avoid flight trajectory

then resume its attack mission once satellite navigation signals are reacquired.

The gravitational potential field function can be modified to

$$U_{att}(q) = \begin{cases} \frac{1}{2}\xi\sigma^2(q, q_g) & \sigma(q, q_g) \leq R \\ \xi\sigma(q, q_g)R - \frac{1}{2}\xi R^2 & \sigma(q, q_g) > R \end{cases} \quad (9)$$

The gravitational force F_{att} it experiences can be expressed as

$$F_{att}(q) = -\nabla U_{att}(q) = \begin{cases} -\xi \sigma(q, q_g) & \sigma(q, q_g) \leq R \\ \xi(q - q_g)R & \sigma(q, q_g) > R \end{cases} \quad (10)$$

where ∇ represents the force acting on the drone in the direction of decreasing potential energy.

To enable drones to quickly escape from threat areas with interference, exponential functions are chosen as repulsion factors. When survival rate is the core consideration, drones can perform evasive maneuvers. On the other hand, to ensure drones can complete attack missions quickly, the repulsive force should not be too large to cause significant changes in the drone's flight direction, so power functions are used as influence factors. When mission completion is the primary focus, drones will not spend excessive time performing evasive maneuvers within interference zones under strong repulsive forces.

The repulsive potential field function generated by the virtual obstacle circle can be expressed as

$$U_{repi}(q) = \begin{cases} \frac{1}{2} [ke^{\eta_1} + (1-k)\eta_2^2] \rho_i^2(q, q_o) & q \in \Omega \\ 0 & q \notin \Omega \end{cases} \quad (11)$$

Among them, η_1 represents the repulsion coefficient when survival rate is the objective, η_2 represents the repulsion coefficient when task completion is the objective, and k is a pre-set weighting coefficient that allows the system to more flexibly balance the target attraction and obstacle repulsion forces, enabling the UAV to demonstrate superior maneuverability in complex environments. The repulsion force F_{repi} generated by the i -th obstacle circle can be expressed as

$$F_{repi}(q) = -\nabla U_{repi}(q) = \begin{cases} -[ke^{\eta_1} + (1-k)\eta_2^2] \rho_i(q, q_o) & q \in \Omega \\ 0 & q \notin \Omega \end{cases} \quad (12)$$

When the UAV is outside the interference zone, the repulsive force is zero, allowing it to fly directly toward the target point. However, when the UAV enters the actual interference zone Ω , the satellite navigation system malfunctions, and the flight control system responds to the repulsive field influence of obstacle circles in the virtual interference zone Ω' , causing the UAV to alter its flight trajectory under the combined effects of repulsive and attractive forces.

Therefore, the resultant force vector F_r of gravitational and repulsive forces acting on the drone can be expressed as

$$F_r(q) = F_{att}(q) + \sum_{i=1}^n F_{repi}(q) \quad (13)$$

The above analysis considers the UAV as a mass point for trajectory analysis at the theoretical level. However, due to its own performance and structural constraints, actual UAVs face various limitations when performing evasive maneuvers. Here, we mainly consider constraints in velocity, acceleration, and turning radius.

$$\begin{cases} v_{\min} \leq \|v\|_2 \leq v_{\max} \\ \|a\|_2 \leq a_{\max} \\ R_t \geq R_{\min} \end{cases} \quad (14)$$

When making turns during flight, drones are constrained by their structural limitations and cannot make small-angle turns. They

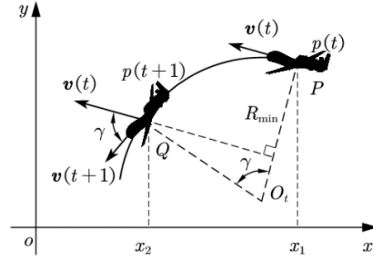


Figure 5: Minimum turning radius

must fly according to their minimum turning radius, as shown in Figure 5.

At time t , let the UAV's position $p(t)$ have coordinates $P(x_1, y_1)$. The UAV flies with minimum turning radius to reach position $p(t+1)$ at time $t+1$ with coordinates $Q(x_2, y_2)$. The rotation angle γ can be obtained as the angle between the velocity direction at time t and the velocity direction at time $t+1$, with O_t being the center of rotation.

Based on geometric relationships, when the UAV turns in different directions, the position of the circle center can be determined by the entry angle β when the UAV enters the interference zone, which can be expressed as

$$O_t = \overrightarrow{OP} + R_{\min} \cdot \frac{1}{\|v\|_2} \text{vsgn}(\sin \beta) \cdot \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (15)$$

where $\text{sgn}(\bullet)$ is the sign function. Then the coordinates of point Q can be expressed as

$$\begin{cases} x_2 = x_{Ot} + R_{\min} \cdot \cos(\delta + \gamma) \\ y_2 = y_{Ot} + R_{\min} \cdot \sin(\delta + \gamma) \end{cases} \quad (16)$$

where δ represents the angle between point P and the center O_t , which can be expressed as

$$\delta = \arctan\left(\frac{y_{Ot} - y_1}{x_{Ot} - x_1}\right) \quad (17)$$

4 Simulation and Analysis

The parameter settings of the UAV and interference source are shown in Table 1, and the simulation results are shown in Figures 6–8

Figure 6 presents the effects of varying the weight coefficient k on UAV survival rate and mission completion rate, as determined by the Monte Carlo method with a simulation time of 500.

The results reveal that k has a significant impact on UAV performance metrics. Specifically, when k is small—particularly at $k=0$ the UAV exits the interference zone quickly, but its survival rate is relatively low. In this case, sufficient time allows for a relatively high mission completion rate once penetration is successful. Conversely, when $k=0.2$, both the survival rate and mission completion rate achieve higher average values; however, their large standard deviations indicate a lack of result stability.

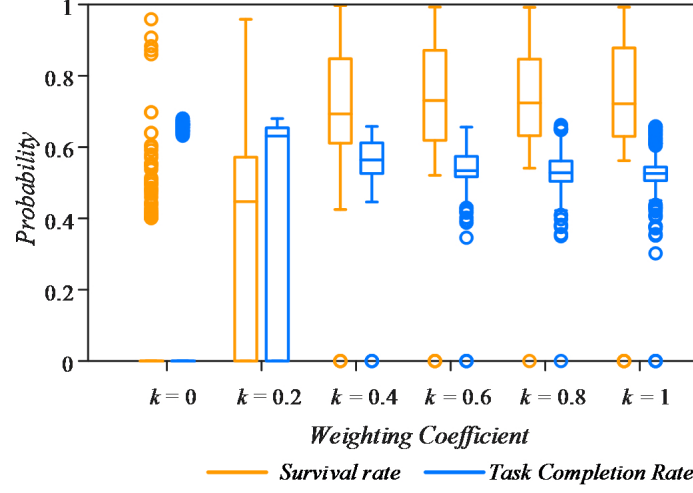


Figure 6: Survival rate and task completion rate at T=500

Table 1: UAV AND INTERFERENCE PARAMETERS

Parameter	Implication	Value
R	Maximum Interference Distance	40000m/s
θ	Interference Beam Width	30°
$a_{\max}$	Maximum acceleration of UAV	8m/s ²
$v_{\max}$	Maximum speed of UAV	150m/s
$v_{\min}$	Minimum speed of UAV	75m/s
$R_{\min}$	Minimum Turn Radius of UAV	2000m
S_T	UAV Survival Threshold	0.4
R_s	UAV Attack Range	4000m
η_1	repulsion coefficient	1.9
η_2	repulsion coefficient	1.3
ξ	gravitational coefficient	0.6

As the weight coefficient increases from 0.4 to 1, the median survival rate gradually increases. In contrast, the task completion rate decreases and converges toward its median value with less variability. These trends suggest that higher weight coefficients lead UAVs to spend more time on evasive maneuvers. Although this enhances survival rates, it may negatively impact overall task completion in time-constrained missions.

Figure 7 and Figure 8 show the effects of different task durations on survival rates and task completion rates when the time is set to 400 and 300, respectively.

From the graph, it can be seen that mission duration has a relatively small impact on UAV survival rate but significantly affects mission completion rate. When the time is set to 300 and 400 respectively, as the weight coefficient increases, the increased repulsive force causes UAVs to spend more time on evasive maneuvers, rapidly reducing the time available for attacks, which leads to a decrease in mission completion probability. The median values are around 0.2 and 0.4 respectively.

Overall, when the current attack target has high value and the mission must be completed at any cost, a more aggressive attack

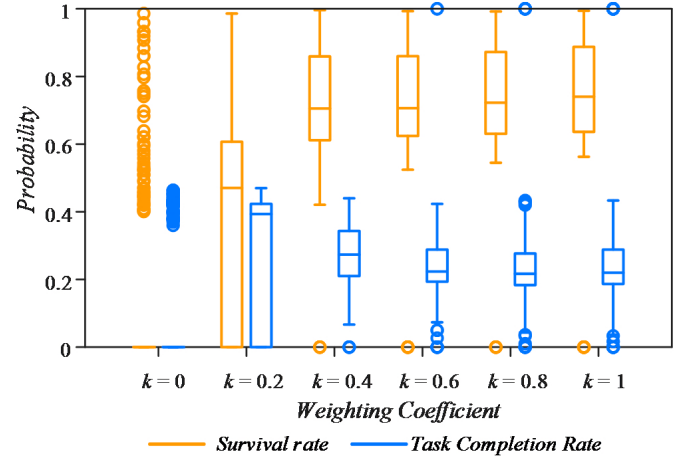


Figure 7: Survival rate and task completion rate at T=300

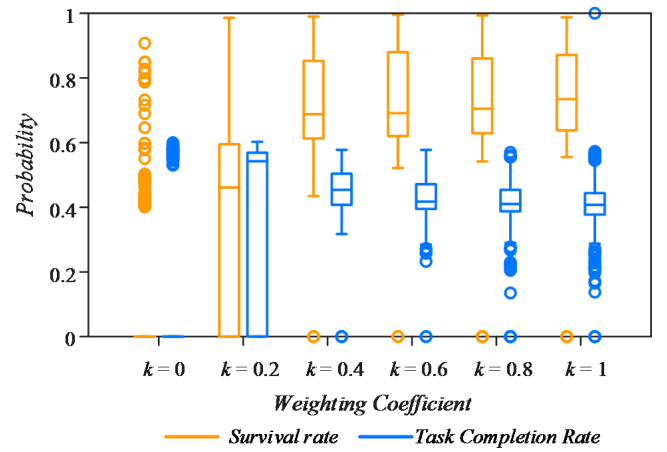


Figure 8: Survival rate and task completion rate at T=400

mode with smaller weights can be considered. However, when the value of the attacking UAV is comparable to the strategic significance of the target, it becomes necessary to maximize the UAV's survival rate. In such cases, a more conservative penetration approach may be adopted, and appropriate weights can be selected based on mission requirements.

5 Conclusion

This paper investigates the problem of UAV penetration under conditions of suppressive electronic interference. By constructing virtual interference points based on the positions of UAVs and interference sources, we generate an effective repulsive potential field for use in the artificial potential field algorithm. Incorporating UAV flight performance constraints, simulation experiments were conducted to validate UAV flight trajectories. The results indicate that the constrained flight trajectories effectively demonstrate different penetration methods when UAVs encounter interference sources. By setting different time parameters to simulate specified mission durations and varying weight coefficients, the study examines UAV survival rates and mission completion rates. The results show that when greater emphasis is placed on mission completion, UAV survival rates decrease, which in turn affects mission completion status. From a comprehensive perspective, setting the weight coefficient around 0.4 achieves a relatively balanced mission completion rate and survival rate, making it suitable for most attack-type UAVs.

References

- [1] Zhang H H, Gan X S, Mao Y, *et al.* Review of UAV Obstacle Avoidance Algorithms. *Aero Weaponry*, 2021, 28(5): 53-63.
- [2] Yan J, Xie H, Zhuang D. The Adaptability of Countermeasures Against UAV Swarms in Typical Mission Scenarios. *Proceedings of 2021 International Conference on Autonomous Unmanned Systems*, 2022: 2436-2447.
- [3] Zhao R, Wang Y, Xiao G, *et al.* A method of path planning for unmanned aerial vehicle based on the hybrid of selfish herd optimizer and particle swarm optimizer. *Applied Intelligence*, 2022, 52(14): 16775-16798.
- [4] Xing D, Zhen Z, Gong H. Offense-defense confrontation decision making for dynamic UAV swarm versus UAV swarm. *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, 2019, 233: 5689-5702.
- [5] Xuan S Z, Ke L J. Swarm Attack-Defense Confrontation Based on Multi-agent Reinforcement Learning. *Advances in Guidance, Navigation and Control*, 2022: 5599-5608.
- [6] Ferreira R, Gaspar J, Souto N, Sebastião, P. Effective GPS Jamming Techniques for UAVs Using Low-Cost SDR Platforms. *2018 Global Wireless Summit*, 2018: 27-32.
- [7] AL-Ghafri, Fat'hi Salim Said, Vidhya, *et al.* Unmanned Aerial Vehicles (UAV) Jammer. *Inventive Communication and Computational Technologies*, 2022: 439-453.
- [8] Fang Y X, Yao Y P, Zhu F, *et al.* Piecewise-potential-field-based path planning method for fixed-wing UAV formation. *Scientific Reports*, 2023, 2234(13): 1-15.
- [9] Wu Z Y, Dong S P, Yuan M, *et al.* Rotate artificial potential field algorithm toward 3D real-time path planning for unmanned aerial vehicle. *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, 2022, 237: 940-955.
- [10] Chen Y B, Luo G C, Mei Y S, *et al.* UAV path planning using artificial potential field method updated by optimal control theory. *International Journal of Systems Science*, 2016, 47: 1407-1420.
- [11] Na Y W, Li, Y L, Chen D Q, *et al.* Optimal Energy Consumption Path Planning for Unmanned Aerial Vehicles Based on Improved Particle Swarm Optimization. *Sustainability*, 2023, 15: 12101.
- [12] Shao S K, Peng Y, He C L, *et al.* Efficient path planning for UAV formation via comprehensively improved particle swarm optimization. *ISA Transactions*, 2020, 97: 415-430.
- [13] Meng X L, Zhu X J, Zhao J. Obstacle Avoidance Path Planning Using the Elite Ant Colony Algorithm for Parameter Optimization of Unmanned Aerial Vehicles. *Arabian Journal for Science and Engineering*, 2022, 48(2): 2261-2275.
- [14] Jie C, Fang Y, Tao J. Path planning under obstacle-avoidance constraints based on ant colony optimization algorithm. *2017 IEEE 17th International Conference on Communication Technology (ICCT)*, 2017: 1434-1438.
- [15] Zhao R X, Wang Y L, Xiao G, *et al.* A method of path planning for unmanned aerial vehicle based on the hybrid of selfish herd optimizer and particle swarm optimizer. *Applied Intelligence*, 2022, 52: 16775-16798.
- [16] Giuseppe A, Kimon P. V, Alessandro R. Fixed-Wing UAV Energy Efficient 3D Path Planning in Cluttered Environments. *Journal of Intelligent and Robotic Systems*, 2022, 105(3):60.
- [17] Lin Z J, Castano L, Mortimer E, Xu H. Fast 3D Collision Avoidance Algorithm for Fixed Wing UAS. *Journal of Intelligent and Robotic Systems*, 2020, 97(3): 577-604.
- [18] Wang Q L, Wu F G, Zheng C C, *et al.* UAV path planning based on optimized artificial potential field method. *Systems Engineering and Electronics*, 2023, 45(5): 1461-1468.
- [19] Sun J B, Liu G L, Tian G H, Zhang J H. Smart Obstacle Avoidance Using a Danger Index for a Dynamic Environment. *Applied Sciences*, 2019, 9: 1589..
- [20] Mah, M C, Lim H S, Tan A W. UAV Relay Flight Path Planning in the Presence of Jamming Signal. *IEEE Access*, 2019, 9: 40913-40924.
- [21] Peng J L, Zhang Z X, Wu Q H, Zhang B. Anti-Jamming Communications in UAV Swarms: A Reinforcement Learning Approach. *IEEE Access*, 2019, 7:180532-180543.